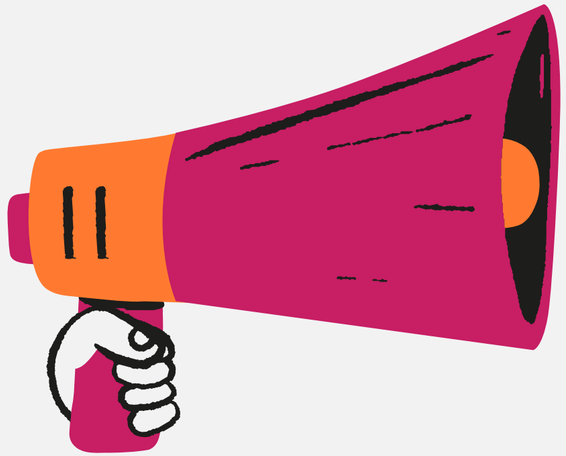
Hospitalised Injuries from Road Crashes in Australia

Team: Group 2

Group members:

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Project Overview



Project Aim

- Analyse hospitalised road crash injuries in Australia.
- Explore trends from 2011 to 2021.
- Compare patterns across states and territories.
- Present findings using interactive visualisations.

Data Source

- Source: BITRE (Bureau of Infrastructure and Transport Research Economics)
- Period: 2011–2021
- Data: Hospitalised injuries from road crashes in Australia.
- Website: Official BITRE data

Progress

- Drafted the introduction and visualisation purpose.
- Identified the official data source.
- Prepared the Design Book structure.

Next Steps

- Complete data processing and exploration.
- Develop interactive visualisations.
- Integrate and review the Design Book.

Data Processing — KNIME

Completed

- Processed three BITRE datasets covering 2011–2021.
- Filtered traffic injuries and calculated annual totals.
- Exported three CSVs: national (11 rows), state/territory (88 rows) and population groups (22 rows).
- Checked outputs against source totals; no missing values or duplicate keys in the exported files.

Challenge resolved

- Some First Nations raw counts were suppressed (n.p.).
- Used the published annual totals instead of treating suppressed values as zero.

Next step

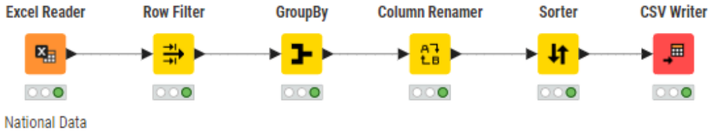
- Use the cleaned CSVs to build and check the website visualisations.

Source: BITRE (2023a, 2023b)

Completed KNIME Workflow

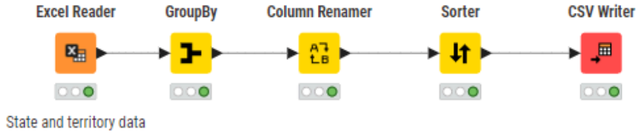
1. National annual hospitalisations (2011–2021)

- Source: National workbook, raw sheet.
- Keep Cause of injury = Traffic.
- Sum Hospitalisations by calendar year.
- Rename columns and sort by Year.
- Output: national_annual_hospitalisations.csv
- 11 rows: one annual total per year.



2. State and territory annual hospitalisations (2011–2021)

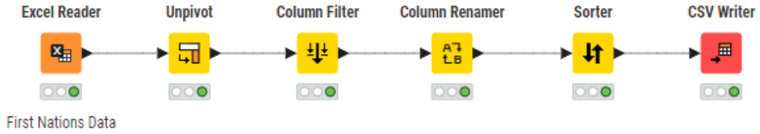
- Source: State and territory workbook, raw!A2:F178.
- Sum the two half-year case counts by year and state.
- The selected counts already exclude deaths in hospital within 30 days.
- Rename columns and sort by Year, then State.
- Output: state_annual_hospitalisations.csv
- 88 rows: 11 years × 8 states and territories.



Data notes: State/territory totals do not sum to the national total, which also includes unrecorded or migratory/offshore residence. Comparability breaks affect Victoria from 2012 and NSW from June 2017.

3. Annual hospitalisations by First Nations status (2011–2021)

- Source: First Nations workbook, T1!A3:C14.
- Use published annual totals because the raw age breakdown contains unpublished values (n.p.) in 2014.
- Unpivot the two population columns and remove generated RowIDs.
- Rename columns and sort by Year, then PopulationGroup.
- Output: first_nations_annual_hospitalisations.csv
- 22 rows: 11 years × 2 population groups.



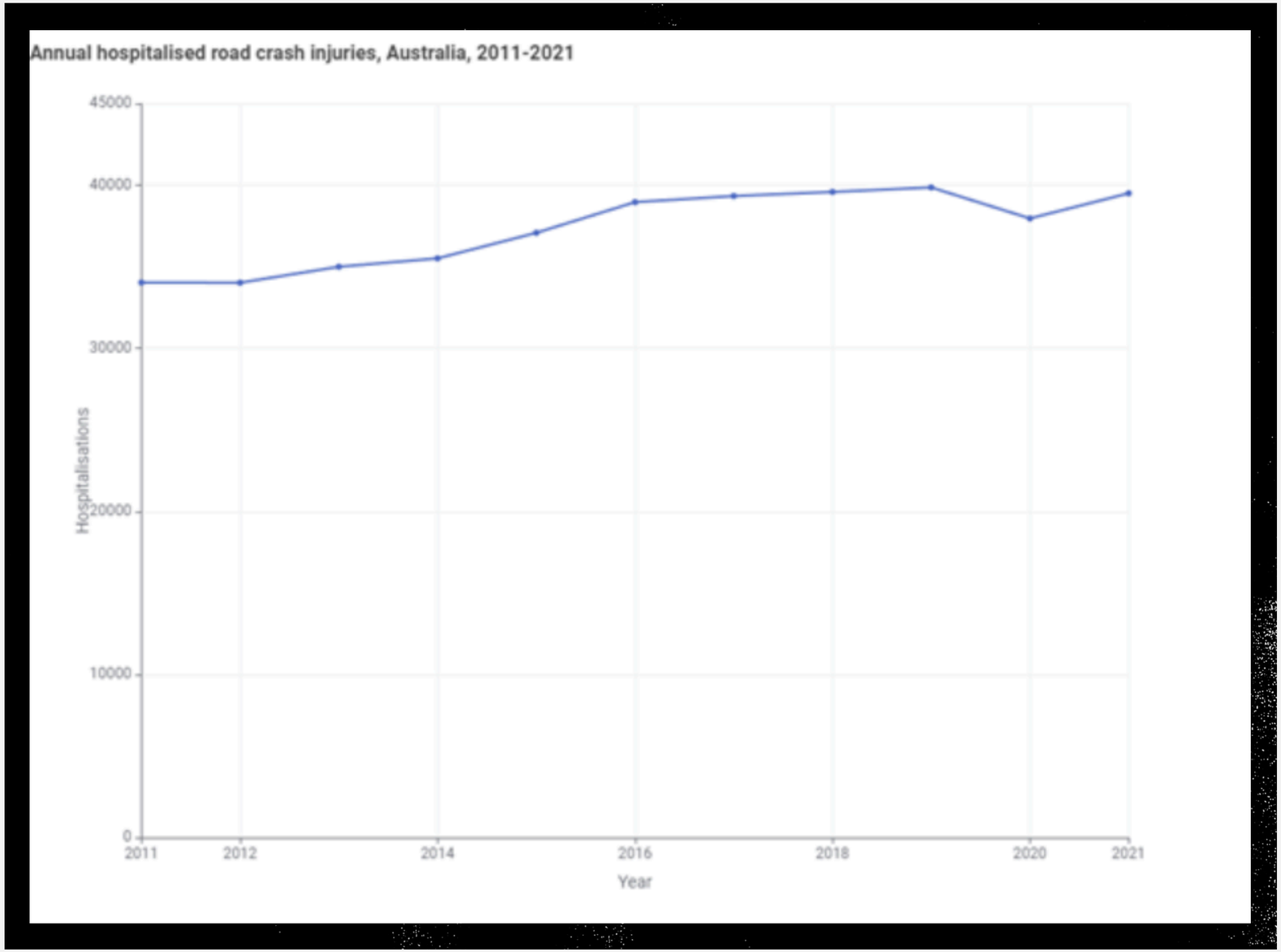
Data note: These population groups exclude cases with unstated or inadequately described First Nations status, so their sum does not equal the national total. Counts are not population rates.

Group 2's KNIME data-processing workflow.

Initial Findings and Charts (National)

- About 34,000 in 2011, rising to about 39,500 in 2021

- Steady increase until 2019 (about 39,800)



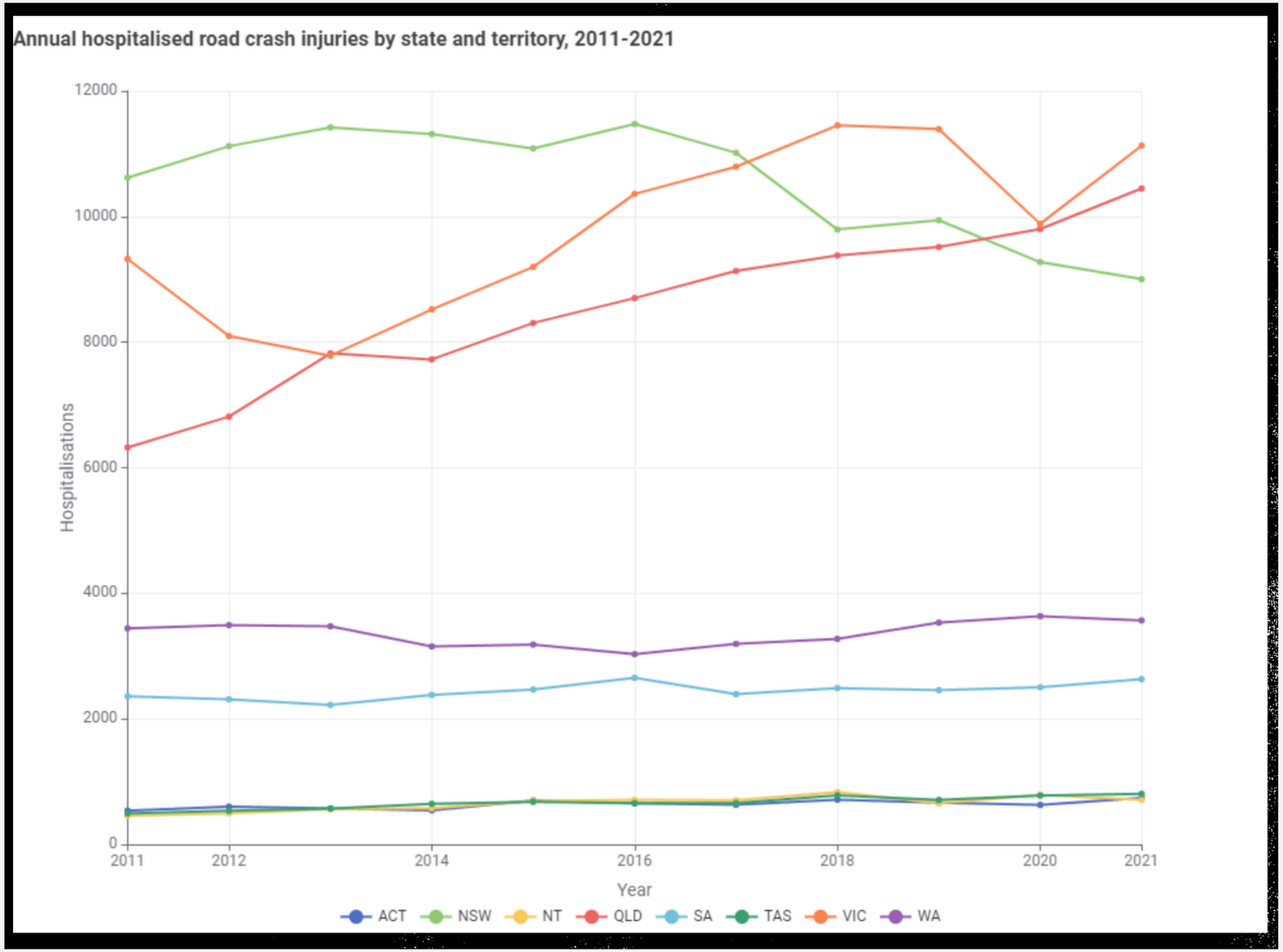
- Dropped in 2020 to about 37,900

- Recovered in 2021

Initial Findings and Charts (state and territory)

- 2021 highest: Victoria (11,130), Queensland (10,447), NSW (9,003)

- Queensland rose the most: 6,322 to 10,447

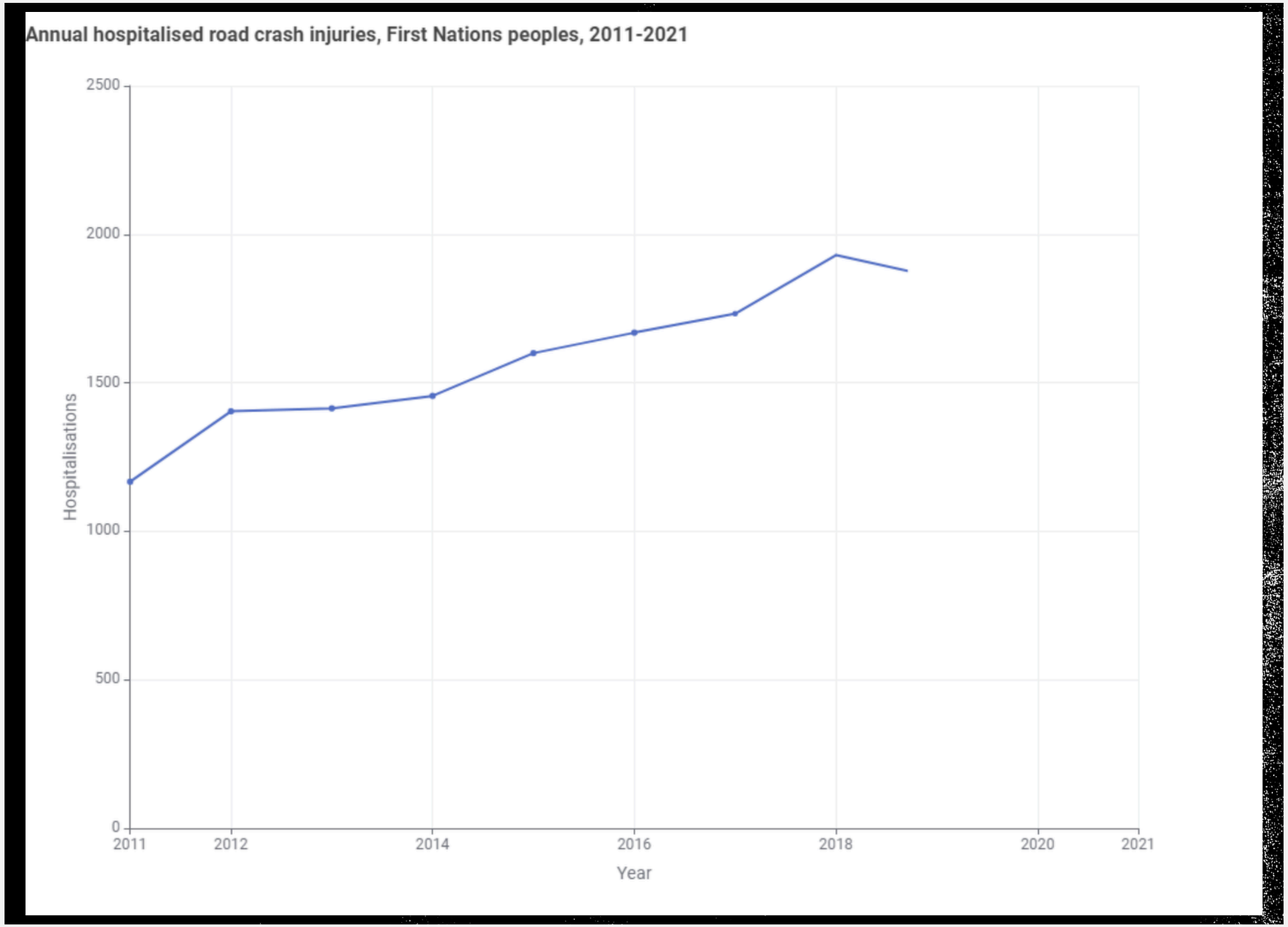


- NSW fell: 10,618 to 9,003

- NT, Tasmania and ACT stayed small

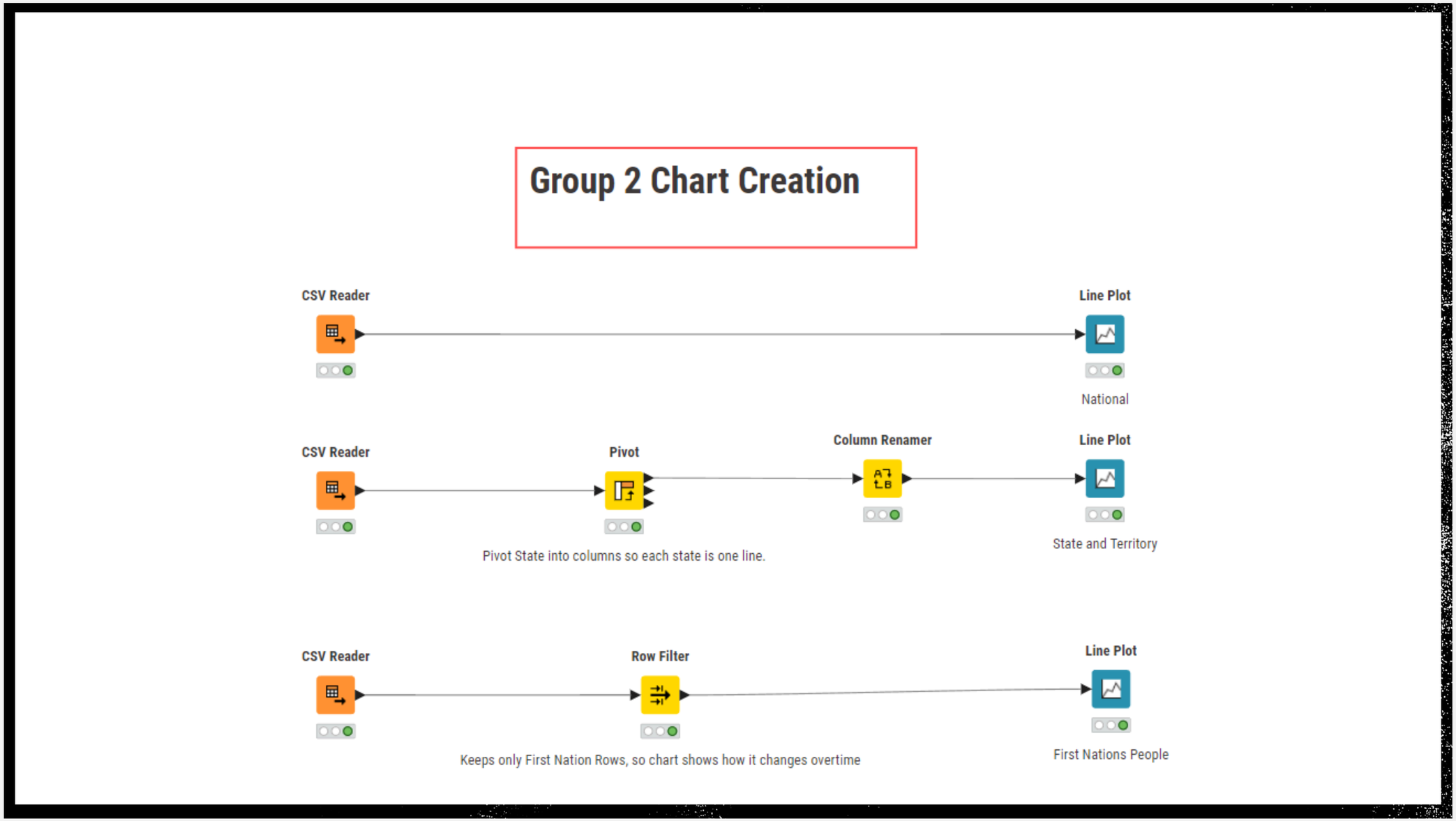
Initial Findings and Charts (First Nations People)

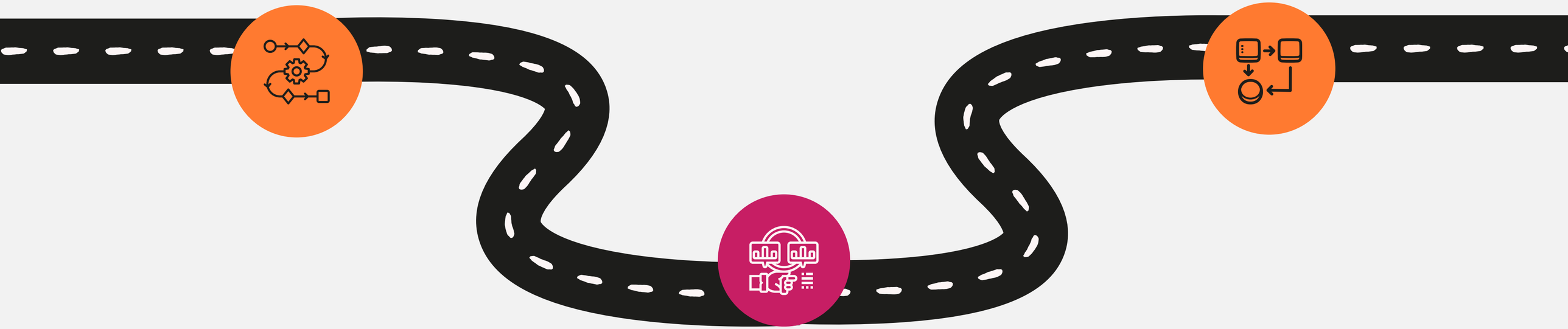
- First Nations: 1,167 (2011) to 2,290 (2021)
- Non-Indigenous: Rose 31,997 to 36,755



- First Nations share: 3.4% to 5.6%
- NT, Tasmania and ACT stayed small
- First Nation Rose again in 2021 while the national count had dropped

How The Chart Was Made (KNIME)





Thank You

